

Step 1: Isolated Theorem Extraction

Theorem 1. (Convergence rate of $\tilde{f}_c$)

Assume that the regularity conditions [R1]-[R3] hold, τ^2 is a constant, and $\gamma > 2$.

As $n \rightarrow \infty$, the prediction in (2) has a convergence rate $V(\hat{f}_c - f) = O_p\{n^{-(1-1/\gamma)}\}$. For $D_k^* \subset \{X_i\}_{i=1}^n$, if $n^{2/\gamma}E(F_X, F_k^*) \rightarrow 0$ as $k, n \rightarrow \infty$, then the prediction $\tilde{f}_c$ in (3) achieves an identical convergence rate of $\tilde{f}_c$:

$$V(\hat{f}_c - f) = O_p\{n^{-(1-1/\gamma)}\}$$

Potential Misinterpretation (if isolated)

- $E(\cdot, \cdot)$ may be parsed as expectation.
- $V(\cdot)$ may be parsed as variance.
- $\hat{f}_c$, $\tilde{f}_c$ and γ are undefined.

Step 2: Global Notation Reference Extraction

Scan preceding sections and collect all explicitly declared notations.

Notation	Definition (Validated)
$E(F_X, F_k^*)$	Energy distance between the true distribution F_X and the empirical distribution of knots F_k^* (not an expectation operator).
$V(\eta)$	Prediction performance metric: $V(\eta) = \int_x \eta(X)^2 dF_X(X)$ (not a variance operator).
$\hat{f}_c$ vs. $\tilde{f}_c$	$\hat{f}_c$: GP prediction using full data. $\tilde{f}_c$: Low-rank predictive process approximation (using D_k^*).
γ	Power law exponent governing eigenvalue decay of the covariance operator (smoothness parameter).

Context Pool

Notation table + local explanations (sliding window)
+ regularity conditions [R1]-[R3] + theorem proof

Step 3: QA Synthesis with Harmonized Context

Question

Notation & Premises:

Let $E(F_X, F_k^*)$ denote the Energy Distance between F_X and F_k^* . Let the performance metric be $V(\eta) = \int_x \eta(X)^2 dF_X(X)$. Let $\tilde{f}_c$ denote the full-data GP prediction and $\tilde{f}_c$ denote the low-rank prediction using rep-points D_k^* . Assume regularity conditions [R1]-[R3] hold, τ^2 is a constant, and the smoothness parameter $\gamma > 2$.

Main Theorem (to prove)

Given that $V(\hat{f}_c - f) = O_p\{n^{-(1-1/\gamma)}\}$ as $n \rightarrow \infty$, prove that if $n^{2/\gamma}E(F_X, F_k^*) \rightarrow 0$ as $k, n \rightarrow \infty$, the low-rank prediction achieves the identical convergence rate:

$$V(\hat{f}_c - f) = O_p\{n^{-(1-1/\gamma)}\}$$

Reference Proof (synthesized):

Uses the definition of $V(\cdot)$ as an L_2 metric and bounds on the Energy Distance $E(\cdot)$ from auxiliary lemmas to control the approximation error between $\hat{f}_c$ and $\tilde{f}_c$, thereby establishing the claimed rate.